Active InferAnt Stream 010.2 ~ plain_text too: Learning Paths, Application Guides, and More Things

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all right welcome to active infan stream 10.2 it's February 23rd 2025 I hope we will stay online long enough through the storm for what should be a very interesting time so thank you everyone for watching keep an eye out in the live chat let's jump right into it all right I'm going to start with a GitHub push and an obsidian push so this is all going to be going to the repo active inference Institute SL cognitive starting the [Music] stream GitHub push then over in obsidian I'm going to do an OB obsidian push so that will be going live to the publish. obsidian DMD slactive inference page and we will select the changed documents and the new 138 documents okay obsidian push going live GitHub push double check it and then let's Jump Right In All right we are in it so this is going to be a little bit wild there's parts that are in various and multiple states of functionality it's just too fun not to share though so wanted to update and follow up on the plain text series and continue to give people some checkpoints and Milestones and open the space for their suggestions and contributions to some very exciting new ways of learning and applying active inference so let's start with the agenda here's the the windows that I have for reference I'm going to put the GitHub window away maybe we'll push later in the Stream there's going to be three main windows up there's going to be a browser where we have the GitHub repo and the obsidian publish format and some other stuff's going to pop up in here later we have the obsidian window here where we can do uh Force directed graphs and this was used also in the previous PL Tech stream then we have cursor 45 running along mostly using agent mode with the Claude 3.5 Sonet so the stream being the second in the plain text series is going to look at learning paths application guides and more things the stream will focus on advancing our understanding and application of active inference and cognitive modeling Concepts through learning paths detailed application guides initially for RX andf for. JL Julia package and a generic things repo with a test Suite that's almost too comprehensive giving some challenges still and other resources we're on

YouTube and elsewhere the information is all linked here and we'll see where code viz comes into play all right we begin with the GitHub push let's check in first with the generic things okay the visualizations that were only minutes prior working just fine of course failed to work but that's totally fine they really are all in there and uh I'm going to pop over to the things generic thing this is seeking and several dozens of prompts and scribbles along the way towards a foundational implementation of activ inference principles through a generic quote thing that can engage in message passing belief updating free energy minimization in a potentially interactive nested Federated situation um it's a little bit of a balance with the markdown files because there's things that look good on the uh markdown display of GitHub and less so in obsidian and vice versa so let's look over to it in obsidian the core idea here as has been suggested and and indexed towards by many over the years is what is that generic active inference thing what is it that we can have that's such a flexible schema that it'll allow us to deal in continuous and discreet time continuous discrete State spaces single agent interacting with the environment multiple agents dissolving agent environment all these different kinds of ideas that are out there and then where PL text is heading and this repo and this Vector of collaborative research is with a lot of the pieces of information be they analytically former formal uh Expressions be they in even probably evocative other plain text representations and eventually multimedia then when we say Implement variational free energy minimization or Implement expected free energy minimization we can know that it's going to use retrieval augmented generation rag like approaches to use that information in the repo to structure the discussion so a lot of the focus over the last several days let's resume it's um just in terms of what cursor is doing in the background um I'm having it run the test Suite again and again while let's see what these ones will not look like um having it run the test Suite in the background in agent mode where it's going to make changes re run the test suite and and so I found that this kind of agent semi-autonomous semi-supervised test Suite operation is super effective um but here's a few of the key elements so I wanted to have the capacity to have the generic thing of course interact with other things this could be Alice and Bob just having that classic conversation that Quantum classic conversation or it could be agent environment or it could be multi-agent situations um here's the concurrency diagram in terms of the message passing now this was a very interesting topic that we'll probably return to when we get to RX infer later is to what extent do we want to schedule the logistics of message passing because as it is with all this General active inference thing research it's like what is the least opinionated schema nothing what is the most opinionated schema well it might be super effective for where it's effectively opinionated but it's not going to have degrees of freedom so which are the degrees of freedom that we really want to engage with when for example reactive message passing in RX and fur which we're going to return to makes some of this message Logistics possibly not necessary for the scheme but again if it's a scheme that's maximally unin ated it's just an empty repo uh I focused on the message passing functionality where starting with a three things that were connected in a linear chain with a lot of test functionality within the message um passing architecture hello Jeff and Monir um here is the logistics of the action perception Loop in active inference again one could remain even more unopinionated on the ordering of the action perception Loop The Ordering of the udal loop and that's the degree of Freedom that RX and fur allows us to relax but here I wanted a langu Independence scheme and just a draft hopefully one where we may actually come to realize that we do want to make the schema already pre-adapted to reactive message passing however if we relax the logistics here so that the reactive message passing can proceed with the least opinionation then that schema wouldn't be able to be implemented outside of a react message passing scheme because it'd be like well where are the logistics and even within a reactive message passing scheme we still do need to have certain kinds of event-based injections in order to get the reactivity going so we have um environment sensory internal and action States external internal sense and action being the blanket States environment passes observations to sensory which passes a belief updating message okay here's the test Suite being emitted so now this is what the autonomous test Suite Improvement looks like in cursor it pops out the test Suite it is passing many of the tests but some of them are still failing and it makes a HTML website that has a lot of useful information a massive fractal test Suite in fact and then just continuing to say all right fix it methodically agentically professionally comprehensively continue to run the test suite and go on and do what you can to fix it and so on those kinds of encouraging fractal prompts so that for each of these core functional elements of the generic thing concept the core which we'll look at free energy processes the inference process Markov blanket depiction the message passing which could be just the standard uh kind of message delivery Pub sub type or it could be more Federated setup being needed in a visualization script here then we have the test Suite in the tests folder which gives these kinds of outputs so it's going to continue debugging while we look through um uh from the internal States expected free energy G across the pi list of possible policies is calculated then action is selected from the policy posterior and then that executed selection uh of action the one hot action gets passed back to the environment so there are other topologies and other ways to schedule the logistics of this message passing but again to the extent that we're less opinionated about some of this stuff we actually have like less and less of a schema so without just having a piece of blank paper or an empty repo empty mind in front of us and say well this is generic active inference how do we put something into that void and more information as with much of procedurally and probabilistically generated code it has all kinds of quirks and features so it isn't High reliability code per se but it's awesome for what it is and it's a starting point for line by line manual investigation and curation and also for further contemporary and F future Advanced code repair strategies um so let's let it continue doing the test Suite it's not giving the visualizations it makes several dozen visualizations I hope that we can get there by the end of the stream somehow someway with several a dozen I know that there's like a kind of see it's a believe it element and certainly the visualizations provide really important visual calibration and gut checks and accessibility for these processes that are happening um so we'll check back in we'll make sure that the um autonomous test Suite it'll stop after 25 agent tool uses I don't think that's needs to be that case but it is the case uh in this version 45 with this current agent but just so we remember ensure that if and where possible we'll we'll let it continue what it's doing now but then it'll eventually say okay that's all I could do right now that visualizations are made prioritized and fully logged with full file name to a terminal okay so we will uh return to the generic thing that that's why starting to debug the generic thing was first on the agenda all right now to learning paths so this is in okay just output another test report this is in docks guides learning paths okay so in the prior plain text 10.1 stream we were having a ton of fun let's look at it in both GitHub I mean we have three ways of looking at this we have the cursor local so this is the code editor version then looking also locally Obsidian and then emulating that file structure of the repo is the GitHub but that won't be updated until the the push is made but we can we can make that push just for fun let's see if there was any visualizations made okay okay okay thing generic thing visualization let's just look at where it get so thing generic thing visualizations okay here we go just to sort of confirm that the visualizations are happening in the debugging so here we go this is the action repertoire with summary as well as precisions on actions and some igen value Spectra on that this is the action repertoire these are belief States and uncertainty these are output as part of the test Suite so that's why these have each time the test Suite is run some of the um time stamping differs other outputs are just without time stamp so they're overwritten here's belief updating not sure how accurately it's implementing basian updating at least though the skeleton and the architecture is there free energy histories here's the free energy landscape from a parameter sweep part of the free energy minimization test looks like it's the same across tests here's the hierarchy some funky stuff happening with the labeling but a parent has child one and child two and then a grandchild here are the uh sensor to processor so that's from sensory to internal and then from internal to action patterns of input and output here's the triadic internal and blanket States for the thing this is a two thing message Network processor see now it's adding more files as this next round of the test Suite continues to occur more belief State distributions correlation heat Maps summary statistics igen values on matrices some very interesting possibilities and hello Ender for spectral analysis on these random matrices uh temporal Evolution this is a little bit of a kind of sudo mandron asking it for full details give the full details the whole repertoire of the sensory the internal and the action States and those can be described as a Through Time process as a uh in time process temporal analysis looking at sensor processor and actuator these three rows and the igen value Evolution test belief updating okay another basian free energy landscape this is a nice looking free energy landscape and roughly Concord with what we would hope to see which is on this manifold where the internal States like -2 and -2 for internal external states where they track each other they have a lower free energy and to the extent that they're off that manifold the free energy is higher and a nice tetrahedral double zigzag me message passing Network okay so we're going to let the autonomous d debugging proceed while we look at the learning paths these learning paths are for people and for non- people so the way this worked is again pop back however many minutes to before where we saw from GitHub that the visualizations were output we had all these different topics across the knowledge base which were more like idea topics just to kind of remind on this in the knowledge base there are these folders so augmented reality and a spatial web and when you are in obsidian the node that you're in sometimes it's a little hard to find in a big Network so here's spatial web and then the what it's linked to are topologically connected and we can drag it around and see what comes along for the ride so there's like that spatial web and then connected to active inference um so these are topics and these these these are like fractal Wiki Pages they could be line by line versioned and annotated and calibrated um as context windows in llms increase and as the efficacy of putting very large knowledge corpuses into finite context Windows also improves relevance realization improves there's really no downside to adding more including a plurality of perspectives on formalisms on qualitative points these are all good and exciting things so in the knowled Bas folder I've added a few dozens of more topics since last uh stream so some of these are let's see they look like they're okay it just sometimes it takes a little bit old Linux laptop doing what it can connecting with a lot of Bucky Fuller stuff which those going through the back catalogs will know has long been a possible promise exciting connection with synergetics intensity and active inference um but these are the knowledge based files here's the mathematics folder so calculus this could be just one very very long several hundred page textbook it could be many textbooks it could be calculus one or it could be calculus adding more adjectives and so on um category Theory same thing complex system sys um and also it can get really specific like desparation which is a Criterion for determining conditional Independence relationships in basian networks so going around and learning through each of these topics is like exploring and super fun and yes needing to be verified and checked and always possible to add but it's like exploring in a lightly synthetic intelligently peer-reviewed Network um in the cognitive folder the approach that I took was to make a file called cognitive phenomena and in obsidian you can choose first off what you want the highlight color to be so in the appearance you can have accent color and it can be any color you like so I like red um the files that are linked and exist are bright red and then there's the dimmed ones which don't exist but a single click will make that file and just drop it into where uh the the top level of the repo so the approach I took with a cognitive folder was to several times iteratively prompt okay let's continue with the test Suite No we we default stop the agent after 25 tool calls please ask the agent to continue manually please continue to run and fix and improve the Suite okay um so several times iteratively agentically methodically comprehensively with adding a little bit of Spice from the sideline throwing in fly balls from left field and all that added in different topics to this cognitive phenomena index file and then said now comprehensively make the files that don't yet exist and so just SP so this can be iteratively done kind of like a two-stroke meta engine where build out the repo uh index add more cognitive phenomena and it can there's no end to how how much specific detail could tesselate down and then the other stroke of the engine is for the articles that don't yet exist make them and for the extent articles improve their dense obsidian style linking and improve their Technical and analytical and applicability and accessibility and so on one note I'll just footnote this is all in English language but there's no reason why it couldn't be on mass translated and snapshotted um so that was the cognitive folder um citations this one is empty really exciting collaborations with the Gen 25 team and others so I look forward to including people and do and documents in in into this as well but for now there's no people and there's no papers um then over the last uh little bit I also added in a lot of Entomology and idology which is B studies related topics and micology which is ant studies so then later on as may very well happen when we say something like make a generic thing specialized for applying to collective intelligence in ants that is very well primed and structured for the llm to be able to make that adjacency and bridge between what we already have about ants and what we already have about generic things so this is in the um there's also some fun topics like mathematical entomology which I don't really even know if I've heard of it done specifically that way but it sounds like it'd be a really exciting class or topic kind of connecting population biology Evolution ecology development spatial modeling cognitive science and behavior experimental design applications like for integrated Pest Management and Biore Regional stewardship and all these other things and then it's like here are five current research applications for mathematical entomology people can have different perspectives on where or how they would value these research applications or how close they are from the uh bench to the bedside or whatever the ento equivalent is but it's like something that can be added to um and then also knowledge base can include specific things like here where we've done a lot of discussion on the bio firm and this is like a documentation fractal about the biofirm addressing the kinds of topics and questions that anyone exploring the biofirm whether they be inner developer membrane Communicator or external somehow relevant to to learn across this it's like how is active inference applied in the biofirm well here's an interactive article on that and and this is all available online like here's what it looks like online here's how the biofirm is doing multiscale nested ecological active inference and applying the free energy principle and then lines can be corrected or improved or added um but again to get from a first pass to here so rapidly is so fun okay so that's the knowledge base now back to the learning paths this was a new piece over the last few days/ weeks so here um still debugging away go get it cursor learning paths again over the years so many discussions well what's the best entry point or and it was sort of like the truth is a pathless land like Krishna mury angle you know what is one entry point and sort of a recognition that there isn't just one there's not just one entry point to anything and so now the interactivity of these tools is actually helping shorten the active inference cycle and time udal Loop timing to an absurd level for live custom generative educational material generation which could be done one shot like you could just go to your favorite llm generative AI chat interface and say I like X please develop me a curriculum for learning active inference so that's something that didn't exist 5 years ago or 10 years ago earlier on in learning active inference but that's something where anyone with access to those chat interfaces can get that one shot but this is opening up with the plain text repo the possibility for having these co-constructed and legible and credible of different kinds learning paths and one person asked in the Discord so the learning path I forget which way they even asked like does it go from active inference to the area or or from the area like well paths go both ways so how octave okay I'm just I think it just dropped for a second okay okay it should be back but let me know if anything in the chat if anything weird happens too many weird things are happening to even say but I think you all know what I mean by that so this active inference economics intersection can be used to take someone from a more generic active inference place or to to the domain or it can be run in the other way I'm going to just drop the bit rate on the stream a little bit right on so um let's this is what it looks like in the local obsidian so here we are and this connects here's the active inference economic learning path let's go over to where it is in so now we're going to look at that I'm going to contrl a contrl k and let's use the grock 2 model comprehensively vastly add relevant sections and Ensure linking Style with exent files okay so here it is running through adding information to these sections we'll see if it adds any new sections as well okay ooh didn't like those parts I I think that in the coming versions of cursor they're going to be improving this um and simplifying their kind of fractal product offering in a way uh unifying the chat and the composer or so I heard from Gary um but we we we can add this it's just that I have it already running in the in the composer with the debugging which maybe we can just sort of we kind of get the picture on the generic thing we get it that we can write Mega fractal modu mod test Suites and we can use that to do this autonomous driven development on different topics and we'll return to it later in the Stream so I'm going to start a new chat with the economics learning path cons comprehensively develop this more we want to use this for a convening that leads to a three quarter co-learning course between an economics department and the active inference Institute so that all backgrounds and familiarities with eg math biology economics Etc are included and have many contributions Avenues so then it's going to be doing this modification in cursor that'll be live represented in my local obsidian and then we can push to GitHub for it to then and and do the shift publish to have it also go live here so then the idea of these learning paths one idea of them is biological learning path like this URL or something very much like this URL is like a stable URL and it can continue to be developed out so instead of where we would be just a couple months ago okay I'm kind of curious about this I really like to do um pottery and embroidery and drawing and this is my favorite you know just like where do you start with active inference well oh well here's the 2022 textbook and here's a textbook group that's awesome that's that's really hopefully exciting that that's not every single Last Mile let alone every last thto meter now there can be a there could be active inference micology learning path so that's from somebody who's studying the generalized active inference topic that can take them out to the mological last miles and from somebody who is working on ants in the field take them in towards this is just one way of studying it um okay here's it just laid it out and we'll say yes make all these additions to the economic learning hello reconfigurability trainer yeah anyone if you have like a suggestion for uh another learning path or a fun thing to try basically we're in the the middle of act two of this stream where we're going to look at the learning paths develop the learning paths a little bit uh then act three is going to go over to RX and fur and explore some parad documentation some fanfic documentation that I made and then uh part four will be wherever the live chat goes okay so here is the modifications that cursor is making to this economics course so again just this is kind of a a micro comment but when we use the control K mode it deleted large sections and then here it even recognizes through the iterative process I noticed some examples were deleted I'm going to add them back in okay so that's happening so now this has been expanded into that multi-order course Concepts okay so that's an example of improving a learning path it could be fractal unrolled into any level of depth it could be we have 15 minutes and it's a ninth grade classroom in this language and here's what they're most excited about or it could be it's a whole Lab and there are five PhD students who are ready to spend 5 years on this topic okay m has made a suggestion this is going to be active inference observation learning pathmd so when wrote human judgment and active especially the observer effect okay I'm not sure if you meant this to be a a learning path but I read it so oh okay and then followed up with this is just a suggestion to see what active inference agent self-perception about its own belief model okay let's turn this into please comprehensively continue to develop these ideas into a learning path it's in the folder with all the other learning paths so it knows a little bit and about the structure and the template and all of that um as a two-way street beep beep for those interested in such topics like as described and observation the art of observation the magic and possibility of observation and active inference parentheses agents and Link obsidian Style accurately but anyone can clone the repo pull the active inference Institute cognitive repo get it going locally navigate over to this folder make a new file we'll do a micology learning path 2 okay but here we're on the observation learning path so it's like kind of like phenomenology kind of like ethology this learning path explores the profound relationship between observation human judgment and active inference it examines how the AC of observation itself shapes our understanding and beliefs while connecting these insights to the mathematical framework of active inference the path integrates perspectives from cognitive science philosophy of mind and computational Neuroscience to build a comprehensive understanding of how agents both human and artificial perceive learn from and influence their environment through observation let's look over at it in the obsidian setting so here's the path okay here it is Little Red Dot here's what comes along for the ride when we forc drag it so let's see foundations of observation phenomenological approaches and scientific interesting inner and outer observation from the Blanket's perspective oh wait active inference Basics free energy principle predictive processing Core Concepts observer effect Quantum measurement Theory social Observer effects active inference and self-perception belief updating and interoception the Art of observation mindfulness and attention Advanced observation techniques oh brother pattern recognition detailed perception contextual awareness Advanced topics computational models applications and implications projects and exercises resources essential reading the mathematics of Mind active inference and agency by Carl friston I don't know if that's a real paper but it's like we're at the point where it's like does it matter because we could generate what would that be okay would you like to add specific content expand the resources yes vastly expand the sections related to exercises there should be so many access ible tangible everyday experiences that will relate to people's direct sensory and cognitive experiences it should be a vast list and one with many exciting starting places for inner outer and social exploration okay meanwhile let's just get the micology one write a comprehensive learning path document for micology we are a team of enthusiastic quote field researchers we are interested in collective intelligence parenthese and also via okay we'll look at that in obsidian in a second and also via nested active inference interested in the nestmate or the colony make the learning path now please thank you okay new chat all right so now while the micology path is unfolding let's look at the observation one thank you manir for the suggestion that was awesome okay now we're looking back to observation all right here are the projects and exercises let's see if any of these strike are immediate fancy inner World exploration sensory awareness exercises one body scanning practice practice I know of one cat that's very interested in that two breath observation series three emotional weather tracking second section of inner World exploration cognitive observation exercises thought pattern recognition memory and perception studies attention Direction experiments wow outer World exploration part one environmental observation projects nature observation series Urban environment studies object Deep dive series wow single object extended study material property investigation function form relationship object lifestyle tracking kind of a nod towards systems engineering pattern recognition exercises visual pattern detection sound pattern analysis time pattern studies social World exploration interpersonal observation projects non-verbal communication study conversation pattern analysis group dynamics observation cultural pattern recognition ritual routine studies media pattern analysis and computational Analysis okay now we return to M mology yes make the learning path math much longer and include EG math diagrams artistic experience possibilities okay while that's improving a micology path let me check the live chat okay reconfigurability wrote what are use cases that this can apply to what do you think looking forward to your thoughts munir thank you again Andrew still catching up from start at 2x speed but you're you're having the 1X experience of the 2x speed the economic and biological learning paths are very illustrative examples what about social science curious how more specific and general impacts and outcomes yeah I mean one answer of course is I would look forward to you adding that file and the GitHub repo uh making that social sciences or we can make a Dub here okay let's let's yes add many ant related topics colon deep lore art culture and folkways amazing historical and accurate Natural History ant anecdotes then we'll look at it in obsidian okay reconfigurability has the topic of underwriting been explored I know that you really like that so let's do a learning path active inference underwriting learning pathmd this is a comprehensive make this a comprehensive learning path to and from the concept of underwriting give special thought and attention to the domain specific ontologies that beginners and experts across the active inference and underwriting domains may bring complexly to such an intersection okay looks good for the ants we'll come back there bring complexity to such an intersection please be vastly comprehensive ontology narrative tools etc for underwriting structure massively okay make this happen agentically comma methodically okay now while it's doing the underwriting learning path let's return to our okay so it's off now this sometimes happens it's off by itself all by its Lonesome but here's how you edit it in obsidian double brackets and then start typing first it suggests it to be um there we go it just took a little bit now we linked it to mycology so now it's a red link because we already have that file micology so now here here we are now here's micology connected to Earth's system schema interesting okay so let's see active inference micology learning path the learning path integrates micology the study of ants with active inference and Collective intelligence Frameworks it is designed for researchers interested in understanding multiscale cognitive processes in ant colonies from Individual ants to Colony level Behavior see this is the Crux of the issue later models and other knowledge niches will not fall into such a trap it's all about what you call the individual individual ants presupposing it as nestmate but it's like okay I'm somewhat um startled I'm surprised from nestm to Colony level behaviors of ants incorporating mathematical computational and artistic perspectives so math computational Core Concepts of micology active inference and social insects research methods field laboratory Advanced multiscale the individual level it's all good it's all good nestmate level single nestmate but now this learning path can be used in generating other materials and it influences and so even though it's an onto linguistic perspective that I have that haphazard usage of the term individual to refer to the nestmate which would be equally inappropriate if it were halfhazard used to describe a colony is what makes the colony's Behavior seen as collective intelligence implicitly supposing that the nestmate is the individual whereas of course the individual quote nestmate or just an estimate is as with all cognitive systems its own composition okay artistic Creative Dimensions let's let's have the um yes please vastly continue to add relevant topics sections exercises train ings reconfigurability to this underwriting document underwriting learning path document okay art scientific illustration technical drawing ant morphology Colony architecture behavioral sequence sketching wow this is very classic environmental context drawing but if you look back in obl and Wilson and all that if not much much earlier this is what was there and some of those are still in use and now with generative AI we could make it look like it was a different species and so on so kind of cool digital art creative documentation photography and videography wow that reminds documentation three art science integration installation art public engagement wow diagram types to master scientific diagrams SD mathematical visualizations artistic representations and then we'll just add in category Theory extended practical projects awesome okay mathematical appendix readings in interesting here the superorganism by hold doin Wilson now this is like a situation where the title of the book falls into the Trap well if the colony is the super organism then that must mean that the nestm is the individual organism and that is the concept that has grasped the imaginations and constrained of many wildly interestingly historically active inference by par pulo frisan at all but it's like here's where the line wise editing matters par pulo friston 2022 MIT press there is also a textbook group about this book at the Active inference it'll take a second to load this at the Active inference Institute tools and equipment additional dynamical systems in Social insect even the concept of calling the study of ants and bees social insects is like oh so then the nonsocial would be the nestmate and then the nestmates are hanging out and that's the composition of the social so it is baked in so deeply into to the language of what is social and solitary insect what is individual and collective intelligence wow and now we can look at it line by line career paths fun okay ooh next steps join the local mological Society everywhere could have one all right let's go to the underwriting all right then this will be the end of this second uh phase then we'll look at RX Ander documentation parad documentation fanfic and then we will go to phase four of the stream with just whatever else people have added in the live chat so thank you reconfigurability for this suggestion and right now intra stream update so I'm going to push that to GitHub then in obsidian shift period I set that it's it you can set your own um keyboard shortcut changed and new 33 new publish so that means that the underwriting what's a mological community yeah reconfigurability I don't know but micology is the study of ants so the mological societies would be like local ant Enthusiast researcher learner chapters okay but let's look at it so now we've pushed it to obsidian publish so let's just look at how it looks in um docks guides learning paths here's underwriting and then now over here here's underwriting so to put this link in the chat reconfigurability there you go all right this learning path provides a comprehensive guide to understanding the intersection of active inference and underwriting designed for practitioners from both domains seeking to bridge these fields some of who might be in the live chat I don't know prerequisites and this is where it could link again more densely um I'm going to have it do this in cursor comprehensively link obsidian style to relevant file names for EG prequisites and everywhere possible so prerequisites probability Theory risk assessment basic machine learning statistical modeling you know again somebody could say well I want there to be a the in the phase space of math and computer science all the quadrants can be explored it could be no math no computer science version could be math but no computers index card style could be computers and no math gen Z Style no I'm just kidding it could be math and computer science together um okay domain the underwriting domain onology this is where the domain experts can say okay what else needs to be added here in terms of the Core Concepts key processes active domain ontology this draws on the models training which is probably also on earlier versions of the otim funology but also um the files okay now it has upgraded all of these so now this is cool so it It upgraded all of these into the obsidian linked format so save that now in obsidian I'm going to do shift period changed here's where the underwriting learning path was changed publish it now wow so cool Incredible offering by obsidian their software and their SYNC feature just it it happened live didn't it so now we can look at it in our local setup here's now what gets dragged but previously it was not connected to anything but now all of these and then these ones that are dimmed these ones could be added but now we oh probability Theory oh here's what probability theory is oh what's a sigma algebra oh well you know that could be linked but back to underwriting um key processes you know each of these could be documented and then fundamental concepts for active inference like yeah what is variational inference oh open that up in a new tab oh it's a method for approximating complex probability distributions through optimization okay okay intersection integration bridging concept risk modeling decision-making and learning systems application areas automated underwriting portfolio management claims processing tools and Technologies technical stack okay learning progression level one understanding basic concepts so underwriting principles again this could be it says the link underwriting principles does not exist but now in here in the relevant location in knowledge base add and link to and from here the quote underwriting principles file make it accurate applicable and more okay so understanding basic concepts of underwriting active inference and risk assessment tool familiarization integration Advanced implementation we could think oh what about multi-agent modeling in this situation just add it foundations of risk assessment now it's getting in another level of detail active inference fundamentals practical implementations okay life insurance underwriting property insurance health insurance oh wow Advanced topics system integration Hands-On risk calculator policy pricer automated underwriting system four to six weeks we could do it in four to six hours portfolio Optimizer that one's tough three to four weeks sounds good certification path level one level two level three industry integration internship program Junior underwriter technical specialist mentorship program continuous learning resources okay now here's the underwriting principles so now I'll push it obsidian here it is in the new guides Concepts underwriting principles okay so it was pushed now remember earlier it said that page doesn't exist and and it didn't but now now it's going to be see where it mentioned it underwriting principles now it's in purple so now here are the underwriting principles utmost good faith insurable interest Indemnity approximate cause and risk pooling wow ubera fides but this could just be semi-manually or fully automated or anywhere in between continue to unfold oh risk assessment we don't have that one now comprehensively add and Link the quote risk assessment file this will be the last part on this and then we'll move to RX infer and then anything else that people suggest okay but here we go risk assessment it doesn't exist okay now it's going to write that uncertainty assessment yeah okay risk assessment written now even though it's probably going to continue to update it we'll push on obsidian so current state we click on that the link does not exist okay so here's the old version of the underwriting learning path now here's one one of the issues um again brought up a little earlier okay new select it publish it published done okay so here current version risk assessment does not exist so it's like we made it en cursor The Prompt was just add and Link risk assessment now reload page this is the the on the I'll put this in the live chat now assessment here's a whole article risk assessment is a systematic process of evaluating potential risks and their impact in underwriting and insurance contexts this foundational concept Bridges traditional underwriting principles with modern active inference approaches to uncertainty quantification and decision-making it totally does wow so that was fun thank you and that was a structured and structuring educational Journey as we explored a little bit about learning and cognitive modeling all right now last section here um for the for the RX andur Learners and practitioners and then section four would you like any other healthy Whimsical fun suggestions all right so here's what happened in the RX in first section okay all right um well RX Ander is a software package being developed by our colleagues and it is updating and developing in very exciting ways we're really happy that we're partnered with them and on this co-learning journey and recently they have done a big update on their documentation as part of the version 4.0 4.01 now release and what's really cool about this documentation site other than the fact that it has an llm integration with Gemini so you can chat with it there but also this whole documentation is a plain text file repo plain text repo in the RX infer uh repository so this docs and this is a very professional software codery way to do this the these documents are a website but also the source code for the documentation not just the code is also available in the repo so what I did is I took their SRC so here's source for SRC right here from I did it within the last couple of days I think and so here's those same files so I just in RX Ander repo online I just downloaded their whole documentation and I dragged it in to docks then we're about to go through some super fun um we you know we'll see what names work you know is it is it um is it Paradox because it's off to the side of the other docs is it heterodox because it's different than their documentation is it fanfic because I'm just a fan and learner and just looking to make some fun documents that help apply and understand RX and fur I don't know but here we are let us go to read me so this is the read me of RX and fur so this is I I with their SRC okay back to here we're in implementations there's a guides implementations the the you know all these folder structures can can be evolved this is all just kind of like in box as we learn some emerging practices for what's happening here um but like we we will do more and different and better so everyone is welcome to learn and contribute and raise issues in GitHub and do more so here it is is the docs here is where I copied in the SRC so they already had this open source repo with documentation I brought it in then over in cursor as always just like reyin the cursor repo just a double check this was when I was doing this and prompted something like here make an obsidian linked read me just to help totally understand RX and fur so here's their summary and that summary exists in our repo too but now here we are with the obsidian linked Style with all these different loopy belief propagation we've been there and then I kept several you know some of these code examples especially it's like let us test them make sure that they work and so on but then and then we'll look through these pages right now because it was like um what's a big Topic in um RX and fur what's a big motivator well doing reactive message passing so here here's the reactive programming data sources reactive layer and then the inference layer Core Concepts stream processing patterns error handling performance considerations these are all plain text these are mermaid mind maps so update frequency we can just add timing and now timing is right there and it's and then when we push to GitHub it'll be updated because this is just a markdown and then when it's published to obsidian sync it'll just be there as part of the diagram message passing in RX infer so just each time I was like huh what would that Wiki article be or what what would that knowledge node be um so let's just look through them we have active inference examples okay and then follow along in red with where okay so here we are active examples were linked out to of course the big heavy active inference one over here that's probably real active inference yep and then a bunch of other ones around here so these are um especially if these can be uh validated and verified that they really run as as we expect them to with the app models that would be awesome but these are just oh what are active inference examples in RX Ander now they have that in the examples in their documentation so if you now we're on their website examples examples. RX in.ml and then here's like the active inference Mountain car so a good cursor a good llm if we said make a new and we'll just let's just do this right now just to test it RX infer Mountain car readme.md okay new chat here write a comprehensively professional read me specifically copying and invoking and building upon RX infers treatment of the Classic mountain car setting use obsidian links densely ensure that all analytical and RX and fur syntax threads are fully unpacked because the the in the docs folder in the examples that it has that full Mountain car it it has the mountain car but now it's going to write an obsidian style read me about the mountain car okay Advanced systems engineering so this was just for systems engineering guide book like the llms are trained on systems engineering documents just like they're trained to on a variety of other technical documents so here's different stages of model design and architecture and life cycle design from a systems engineering perspective and then this is the advanced systems engineering guide book I guess um okay let's look at the mountain car here's the RX infer Mountain car read me okay that part could some stuff again it renders really well in GitHub but not in a local obsidian so so you know but then that's where we get the metad docs that's like we like this kind of linking and that kind if it's called this or if it has this template we like that if it's like that we like this okay so we could check does this this looks pretty similar does that actually bring in oh probabilistic model okay and then it's like we already have that page but if we didn't we could say write that page go back inference constraints free energy components active inference framework optimize the expected free energy let's link active inference there double bracket inference so now it's a red link because we already have that page policy selection oh what is policy selection it's like middle click how do we think about policy selection in active inference well we use expected free energy what are the expected free energy components well one decompositions epistemic and pragmatic value what variabl is used to apply that what notation or what letters it associated with sometimes with the E and so okay and then here's the running like this just generated this obsidian Style plus plus plus from the veritical mountain car example that the developers provided um yes add many more sections as relevant including an intro section that sets up a wonderful story as to why we're solving it this way okay back to the RX and fer dos so compiler pipeline pending experts and others questions and comments but this is very fascinating to look at how it digs into the documentation and the code itself and pulls out some topics that even in the year plus of learning about this package we haven't necessarily gone too much into detail but this could help a lot with automated code production and Improvement and also onboarding and and better understanding of performance all these different features so that's the compiler pipeline code Generation all right accept it back to the mountain car imagine standing at the bottom of a steep valley with an underpowered car your goal reach the summit the catch your engine alone isn't strong enough to climb directly this is the Classic mountain car problem but today we're going to solve it in a fundamentally different way not through traditional reinforcement or optimal control but through the lens of active inference why active inference and it's like and then the show is on the road okay um execution engine of RX and fur here's what it looks like over GitHub that top section looks like that in the obsidian program locally but it this on GitHub which is epic execution engine message flow architecture optimization flow m message passing message scheduling cool Factor graphs key topic so this is a display kind that doesn't work in the GitHub but let's look at it over here let's see if it looks fine okay this one has an error let's see if it can fix it Factor graphs we are are getting some diagram and markdown weirdness and failures to render properly on GitHub please expand improve obsidian style link and ensure total comprehensive representation of RX infer technical contents within properly appearing diagrams and Etc okay free energy how is free energy playing a role in RX and fur how is it calculated that's what this article is and then it's like well this is like very technical that's a good thing it serves as a concise linkable place and also it can be entered into with learning paths write me a learning path to get to free energy in policy inference as planning in RX infer for an active inference agent and I'm coming from this perspective how does free energy play a role in message passing here's perception belief updating with variational free energy and then action in with expected free energy and where they come together that's epic message passing in active inference some of this could be lowkey fabricated but having the RX inferred docs themselves there gives a scaffold and if it was ambiguous enough to um need so much interpretation that's a sign to improve the underlying documents and to improve Paradox as well um more details this one I just said write write a followup give more details on how free energy plays a role in message passing okay here okay it made the improvements to the factor graphs article let's return to factor graphs this part still looks like it's an error let's see if it saved it push it to GitHub okay so unable to render R display reload it new version just happened now still doesn't work but people kind of oh wow that's so cool that's Factor graph style um getting started how do you get started in RX and fur well here's conceptual PR Rex here's the installation this looks good basic concepts your first model coin toss that's what we did too visualizing using plots more advanced features working with constraints custom distributions this is getting into some pretty Advanced examples um message passing the concept model macro this is the at model macro Paradigm this is because this is kind of the core interesting syntax that opens up an incredible world of semantics the at model macro automatically converts model specifications into Factor graphs it uses this app model Style syntactically to develop these enormously performant reactive message passing setups which can be used for batch learning or for streaming models can have constraints applied in any which way and uh enables reactivity to occur with this kind of subscription based approach as that we could explore more about that add another comprehensive RX infer documentation page fully linked about a abstract syntax trees it's kind of an Arcane topic but it matters model specification here's how you specify a generative model in RX and fur here's fanfiction in interactive documentation about how you do that um RX and first stack these are some visualizations that I don't even know if they've existed before new ways and not necessarily even accurate ones but sometimes on the path towards utility and learning of the model specification through the app model macro here's the as transformation that's the abstract syntax tree we're about to look at that page that is being written right now yes add vast comprehensive sections to a page with analytical details diagrams connections okay so the app model is specified syntax Wise It's transformed using this abstract syntax tree from that as transformation The Factor graph is compiled rules for message passing are compiled optimization passes are performed we could look in more detail what are those then with the message rules just in time prescribed and the optimization passes of some kind we could explore situations where they are and aren't coming into play then there's this execution engine that actually runs on those rails as constrained and specified through message passing node local message passing free energy minimization and there's some memory management too okay what is a parses the code extracts the variables identifies the dependencies generates a graph optimization eliminates dead codes merges optimizations and optimizes memory now we'll accept the as article let's jump over to this new as processing article abstract syntax trees play a crucial role in model specification within ARX andur the app model macro transforms Julia code into an as which is then processed to generate Factor graphs for probabilistic inference a type system is based on type Theory principles here's type theoretic treatment of as processing here's how it connects to Julia type abstract distribution continuous or discreet two examples of continuous distributions normal gaussian and the and the gamma message passing on a memory management in a and here's how a variational inference is done in our for here's streaming inference in our actioner okay um let's look at one quick code viz okay meanwhile um that completes our primary agenda let's reload this window um at this time if anyone has any requests or ideas write and I will stick around for a few more minutes otherwise I hope this has been interesting useful possibly relevant um let's so here um code viz is is doing its own indexing all right let's do okay I'll just make one okay thank you reconfigurability okay munir has written one suggestion I'm just going to show how that how I'm doing that there's I I know more and better and different and relevant for you um okay just like wait a minute like where is everything okay this is going to be active inference okay and then um we'll just call that um path for for a second when wrote I have another topic experience as a method by Carl popper and the least action principle free energy minimization theory of mind I'm curious to see the output as a learning path all right we'll call this experience as a method or the paparian learning path inference experience do we want to make this like Jimmy Hendrick style experience method path develop this learning path with the stated topics comprehensively including some Jimmy hendris like St ings quote are you experienced okay check out code viz okay still the code VI sometimes takes a while I know that they're aware of that let's just see if it gets there by the end of the Stream Okay add comprehensively obsidian linking to extent files and Carl popper philosophy everywhere possible pragmatism learning pathmd this is kind of fun here's just doing tab autocomplete style and you can get do we want to complete that one I don't know if it's the time but we'll okay so let's let's do paparian okay that's the experience method one here in the pragmatism path right vastly and comprehensively related to American pragmatism and operationalism okay let's look at the areu experienced learning path are you experienced a journey through active inference and scientific method knowledge speaks but wisdom listens Jimmy Hendrick I may be wrong and you may be right and by an effort we may get nearer to the truth Carl popper okay experience as method walking through the Purple Haze of knowledge Carl Popper's epistemological Revolution practical experiments problem tenative solution error elimination new problem yes add more information on the history of American pragmatism then that we're going to CU up this next one returning to the experience method add a lengthy section contrasting the paperian hendris epistemology with the hegelian synthetic dialectic method and cont okay we'll return to that one after that's let's do little pragmatism metaphysical Club absolutely Cambridge Beginnings Holmes Wright James parce Emerson Classical period 1878 to 1920 Pierce theory of science James the Golden Age 1920 to 1940 mid-century transitions logical empiricism contemporary Renaissance interesting foundations of thought operationalism and scientific practice Bridgeman indeed it really was that way whoa Neo operationalism cognitive operations I'm sure cognitive security otim through the pragmatic lens social Dimension so cool so cool okay okay let's go back to the experience as method see if see if that hegelian contian element has a Jimmy Hendrick song that that reflects it okay philosophical Crossroads bold as love critical as truth are you experienced have you ever been experienced well I have Jimmy Hendrick experience is not what happens to you it's what you do with what happens to you Carl popper wow hegelian dialectic the Purple Haze of spirit Conan Foundation transcendental structures paparian Revolution electric experience synthesis and contrast the wind cries method Wind Cries Mary integration without AC inference All Along the Watchtower contian categories predictive models hegelian development hierarchical learning preparing and testing error correction the story of life is quicker than the wink of an eye Jimmy Hendrick the growth of knowledge proceeds from old problems to new problems Carl popper remember like hendrix's Mastery of the guitar and Popper's Pursuit Of Truth understanding these principles comes through practice experimentation and bold exploration let your mind be free to experience predict and learn while maintaining critical thinking okay here we'll just do one final one ensure that this active inference so I'm going to type this out generate it probably read it if it looks funny and interesting then I'll read any final comments in live chat and then that will be the end ensure that this active inference learning path weaves delightfully around the who's on first famous dialogue the topics of active inference and fre energy principle should comprehensively and technically conveyed amidst a rarified soup of baseball lore Timeless comedy wonderful wonderful science and the occasional appearance by the ghost of John Nash what wonderful dialogue intergenerational family Dynamics and deep pH okay that sounds good deep philosophical questions of identity agency and the nature of of reality as interpreted in a question question 20's golden quote golden era of American culture in technical terms that are accessible to a wide audience ensure detailed baseball lore and active inference all along the Every Way new chat make this happen structure the learning path itself as who's on first not just in semantic reference all right code VI so you still gather 11 minutes in still frozen so what is this code base this is a large repo so I kind of get it vastly expand vastly expanding this learning path ensure technical math sections and topics and diagrams and obsidian linking and long long long conversation conveying so much lore and and information comma it really does Boggle mindes yeah cursor indexing works really well code viz not quite there for large databases yet okay thank you Jeff thank you Andrew Monir yeah we'll see where this gets us if it's like funny and interesting possibly read it then that'll be it welcome to a unique okay active inference learning path who's on first introduction the perfect game of Mind welcome to a unique EXP exploration of active inference and the free energy principle structured as the immortal Who's on First comedy routine just as abbid and Costello's routine plays with language and expectations will use their format to unpack the fascinating world of how brains make reality okay there we are for for those who like a little topological side with their baseball comedy all right right okay I don't think I'm going to read this whole thing but it's clearly funny let's just see what happens act one tell me about active inference okay brains predictions technical interlude first base you're saying there's someone called prediction on first who's on first I'm asking you who's making the predictions Deep dive second base strategic play historical baseball analogy M just as Bill James revolutionized baseball with saber metrics Carl frisan transformed Neuroscience with a free energy principle that should be an obsidian link okay whoa sensory input prediction who free energy what action I don't know it's like the UDA Loop but for comedy but for the 19 1920s but for baseball but for future audiences Third Base action selection The Outfield extended Concepts okay little bit of little bit of dialogue little bit of chatter little bit of peanut gallery practice drills who's making predictions predictive processing exercise what's the free energy mathematical workshops I don't know is on policy selection here we go historical context the 1920s Golden Era baseball statistics meets brain statistics that is super cool add many more connections about baseball statistics and brain statistics give PhD dissertation level and length about how statistical parametric mapping SPM for sages even the future of advanced analytics SPM could be you know posterior probability maps for where the ball is pitched or hit okay baseball statistics meet brain statistics Babe Ruth's batting average as a probability distribution Miller Huggin strategy as policy selection and the 1927 Yankees as a predictive processing machine cultural integration Advanced topics the perfect game statistical Bullpen notes for Learners okay W see it just changed while we're in it okay now here's the statistical SPM meet saber metrics hey Abbot these numbers are making my head spin that's because you're looking at them like box scores you need to think about them like brain Maps gum random field theory in baseball and brains it it literally is that way Advanced analytics the convergence spatial normalization and player normalization literally it is that way temporal basis functions pitch sequencing effects fatigue modeling statistical methods it's like I feel like people are ready for SPM applied to baseball but baseball applied to SPM is Evergreen okay I'll read this last comment and then any other live chats and then it's over remember just as abbid and cella's routine seems confusing at first but has a clear internal logic active inference might initially appear complex but follows coherent principles keep practicing and soon you'll be running the bases of cognitive science like a pro ending of number 0 10.2 push the obsidian push the new and the updated done close obsidian okay ending of 10.2 close it thank you Claude and so many ways we can continue from there but that's where I'll call it for tonight so I'm glad that the internet held up mostly well enough to make this happen looking forward to where people take it all from here so till next time peace for